Question 1 - Executive Diarization Benchmark Report

Public Evaluation Standard for Conversational Indian Speech - Executive Briefing

Core Strategic Objective

Enterprise conversational AI, customer QA, and meeting platforms require robust speaker diarization ('who spoke when'). Single aggregate scores fail because production systems face diverse trade-offs across overlap handling, reverberant hardware, latency, and cost. This benchmark establishes a transparent, reproducible 100-hour evaluation standard specifically targeted at Indian conversational speech.

Committed Primary Scoring Standard

- **Primary Metric:** Diarization Error Rate (DER) evaluated at **0.0 s collar (zero-collar)** with **overlap speech included**.
- **Speaker Mapping:** Optimal Hungarian bijection per session; no cross-session identity leakage.
- **Speaker Cardinality:** Autonomous estimation; oracle speaker counts strictly forbidden on primary leaderboard.
- **Statistical Rigor:** Paired conversation-level bootstrap 95% confidence intervals; overlapping CIs declared ties.

Corpus Allocation: Exactly 100.0 Hours

Held-out dataset of 250 conversations across 4 stratified cohorts: (1) 40.0 hrs Indian English (1-8 spk, 50% in-person/50% online); (2) 25.0 hrs Hindi & Hinglish code-switching; (3) 20.0 hrs Regional code-switching (Tamil, Telugu, Marathi, Bengali); and (4) 15.0 hrs Telephony & noise stress cases (8 kHz PSTN, reverberant halls, cafeteria ambient).

Evaluation Shortlist: 10 Systems Across Deployment Paradigms

System	Provider	Model Checkpoint	Deployment	Key Evaluation Focus
pyannote.audio	pyannote	speaker-diarization-3.1	Self-hosted	Open-source neural baseline; native overlap
NVIDIA NeMo	NVIDIA	MSDD / Sortformer	Self-hosted	Production GPU streaming & batch diarization
Deepgram	Deepgram	Nova-3 (diarize=true)	API	Fast turnaround speech API diarization
AssemblyAI	AssemblyAI	Universal-2 + Diarize	API	Commercial batch transcription & speaker labels
Google Cloud	Google	Chirp 2 (diarize)	API	Multilingual foundation model diarization
Azure AI Speech	Microsoft	ConversationTranscriber	API	Enterprise meeting transcription & turn attribution
AWS Transcribe	Amazon	ShowSpeakerLabels	API	Asynchronous speaker clustering in batch STT
Speechmatics	Speechmatics	Enhanced Diarization	API	High-accuracy multilingual turn alignment
Gladia	Gladia	Whisper-based Diarize	API	Real-world audio transcription with speaker ID
ElevenLabs	ElevenLabs	Scribe v2	API	Speech-to-text with turn boundary attribution

Enterprise Buyer Decision Framework

The public leaderboard provides 10 supporting metrics: Missed Speech %, False Alarm %, Speaker Confusion %, Overlap DER %, Speaker Count Error (SCE), Speech Coverage %, Real-Time Factor (RTF), Latency (s), Cost/hour, and Failure Rate %. This multi-metric architecture empowers buyers to select systems optimized for their precise acoustic environment and SLA constraints.